FRANCESCO PISACANE

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EDUCATION

Politecnico di Milano *Sept 2024 - Present*
MSc in Mechatronics and Robotics, Mechanical Engineering (GPA: 29.9/30)
– ***"Ing. Roberto Rocca" Scholarship, Top 6 MSc Engineering students*** (€2.800)
– ***"Ing. Aurelio Beltrami" Scholarship, Top 10 MSc Engineering students*** (€2.800)

Ghent University *Sept 2025 - Jan 2026*
Exchange programme in Electromechanical Engineering

Politecnico di Milano *Sept 2021 - Jul 2024*
BSc in Automation and Control Engineering
– ***"Ing. Aurelio Beltrami" Scholarship, Top 9 BSc Engineering students*** (€2.800)

EXPERIENCE

NASA Jet Propulsion Laboratory (Pasadena, CA, USA) *March 2026 - Sept 2026*

Selected for a 7-month research period to develop the Master's Thesis on Modeling, Dynamics, and Control of Distributed Planetary Vehicles. Research funded by the ***"Carlo Sacchi Future Talent Award" Scholarship*** (€20.000)

PROJECTS (RELEVANT TO THE ROLE)

**Additional projects are available on my GitHub profile: https://github.com/FrancescoPisacane*

Multi-Agent Reinforcement Learning (MARL) for Swarm Coordination
Developed a decentralized RL framework for multi-agent coordination in dynamic environments.

SLAM and Computer Vision for Autonomous Object Search
Developed a ROS2-based Python architecture integrating SLAM, navigation, and object recognition with YOLO for real-time visual detection.

PPO Reinforcement Learning for Electric Battery Optimization
Developed in Python a Proximal Policy Optimization framework for optimizing energy scheduling in a photovoltaic system, reducing electricity costs under dynamic pricing.

Bayesian Optimization for PID Tuning in a Robotic Arm
Implemented in MATLAB a Bayesian Optimization routine to tune PID gains and estimate link masses in a 7-DOF robotic arm; achieved 10^{-4} relative error via problem decomposition based on system dynamics.

EXTRACURRICULAR ACTIVITIES

IDEA League Challenge Programme (2025 – 2026), joined through a merit-based scholarship. International honours programme with study periods at ETH Zurich, TU Delft, RWTH Aachen and Chalmers University.

Automation Engineering Association (AEA): Project team member (since 2021).

Summer School in Applied Geophysics, ETH Zurich (2025), joined through a merit-based scholarship.

LeadTheFuture Mentorship Programme (since Sep 2025), a highly selective STEM mentorship (<13% acceptance) with mentors from leading innovation hubs like Silicon Valley and CERN.

Junior Enterprise Milano Politecnico (JEMP): Audit team member (since 2024).

Aerial Photography with Drones: Independently designed and captured aerial videos for real estate agencies using a custom-built drone.

TECHNICAL SKILLS

Programming, Simulation and Tools: MATLAB, Simulink, Python, C, ROS2, Gazebo, Docker, PyTorch, Git, Linux

Methods: Control Systems Design, Digital Signal Processing, System Identification, Experimental Modal Analysis, Bayesian Optimization, SLAM, State Estimation, Path Planning, Deep Neural Networks, Reinforcement Learning, Model Order Reduction

Languages: Italian (native), English (IELTS 7.5, C1)